I cleaned the uploaded content into **proper interview questions**, grouped by topic. I treated vague entries like “op-amp questions” or “resume questions” as real prompts by rewriting them into clean question form, and I kept unclear/non-disclosed entries separately so nothing silently disappears. Source content includes the raw Hardware Test Engineer question dump and the previously consolidated Glassdoor extraction.

Hardware Test Engineer Interview Questions — Categorized & Cleaned

1. Hardware Test, Validation, Debugging & Failure Analysis

1. How would you test a RAM?
2. How would you test hardware from a software and hardware point of view?
3. Several clients set up a product at their facilities. It fails at some sites but not others. How would you debug it?
4. How would you test a product more efficiently?
5. Given a subassembly, how would you design a fixture to test it?
6. What is your plan to test the hardware?
7. How would you go about testing an unknown product or system?
8. How would you test a buffer using an oscilloscope and other instruments?
9. What practical testing problems might you expect, and how would you debug them?
10. Describe your method for creating a test plan.
11. What are your thoughts on Design for Testability when designing products?
12. How would you develop test plans?
13. What is the difference between shock testing and drop testing?
14. Give examples of when shock testing and drop testing should be used.

15. Given a completely unknown PCB, how would you test it?
16. If there are n inputs and m outputs, how many patterns are needed to test a circuit completely?
17. Explain stuck-at fault models.
18. How would you detect whether a wire is stuck at 0?
19. Do you have test-case development experience?
20. How do you pick electronic components for testing?
21. How would you test a motor driver?
22. What would you do when a chip does not work?
23. How would you design experiments to test for failure?
24. What are the best bugs you have found?
25. How would you calibrate an ADC or DAC?
26. What happens when an inductor in a DC-DC converter saturates?
27. How much PCB testing experience do you have?
28. How would you debug a physical board?
29. How would you characterize the power efficiency of a power supply?
30. How would you detect a hardware fault in memory using C?
31. How would you verify a full chip?
32. How would you test lab equipment and prior testing setups?
33. Build a circuit using components, a breadboard, DMM, oscilloscope, and datasheet.
34. How would you research and approach a specific project?
35. How would you test for failure in an experiment?
36. What lab testing questions have you encountered?
37. How would you characterize a failed hardware part?

38. What test-design considerations would you use for a regular water bottle?
39. How would you design a fixture or test setup for a hardware subassembly?
40. How would you measure very small currents, such as nanoampere current from a photodiode?
41. What are different current measurement methods, and what are their pros and cons?
42. How would you measure PSRR?
43. How would you read an ADC value from a microcontroller?
44. How would you reduce noise in sensor readings?
45. How would you remove crosstalk between channels in a CMOS image sensor?
46. How would you characterize a power supply using lab equipment?
47. What are the voltage waveforms in the given circuit?
48. Analyze a provided circuit and explain how you would test it.
49. Connect a circuit and explain its operation.
50. Describe how experiments should be designed to isolate hardware failures.

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2. Analog Circuits, Mixed-Signal & Circuit Theory

1. Explain basic analog circuit knowledge.
2. Explain basic circuit theory.
3. Analyze a basic circuit diagram.
4. Analyze amplifier and circuit diagrams.
5. Explain circuit analysis.
6. Answer questions about circuits, op-amps, and transistors.
7. Draw a basic power circuit and an analog circuit.
8. Explain how a peak detector works.

9. Draw op-amps in inverting and non-inverting configurations.
10. Draw an op-amp circuit and derive the gain formula.
11. What is the input impedance and output impedance of an ideal op-amp?
12. Identify the op-amp configuration in a given circuit.
13. Design an LED driver using an op-amp.
14. Design a circuit to produce triangle waves.
15. Design a low-pass filter.
16. Explain a low-pass filter.
17. Explain different types of filters and their characteristics.
18. Explain the operation of a wind-noise rejection filter.
19. Design the wind-noise rejection filter as either a DSP hardware block or an analog hardware component.
20. Draw the schematic of the wind-noise rejection filter and identify its poles and zeros.
21. Explain RLC circuits.
22. Explain Bode plots of RLC circuit filters.
23. What is the behavior of a series LC circuit and a parallel LC circuit?
24. What is the resistance between two opposite corners of a cube where each edge has resistance R?
25. What is the voltage-divider output when $V_{in} = 12$ V and $R1 = R2$?
26. What is the power dissipated when current is 2 A and resistance is 100 Ω?
27. Find the reactance when capacitance is 10 nF and frequency is 100 kHz.
28. How does a capacitor work?
29. What is the use of a pull-up resistor?
30. Where should you place the capacitor on a power pin?

31. Draw the voltage waveform across a capacitor with two switches in parallel with two capacitors.
32. Explain the voltage relationship when a voltage is applied across a capacitor and resistor.
33. If a capacitor is added at the base of a common-emitter BJT, how does it change the function of the BJT?
34. Explain MOSFETs.
35. Explain BJTs.
36. What are the three stages of operation of a transistor?
37. What is a MOSFET?
38. Explain feedback amplifiers.
39. Explain comparators.
40. Explain capacitance sensing methods.
41. Explain basic amplifier operation.
42. Analyze panel circuits.
43. Explain audio fundamentals about amplifiers and measurements.
44. Explain basic electronics concepts.
45. Explain basic EE concepts.
46. Explain voltage regulation.
47. What is a voltage regulator?
48. What are the pros and cons of different voltage regulators?
49. Explain the difference between AC coupling and DC coupling.
50. Explain analog design fundamentals.
51. Explain digital design fundamentals.
52. Explain op-amp-related questions.
53. Explain transistor basics and perform quantitative analysis.

54. What happens in a circuit when voltage and current delivery change?
55. What is the relationship between voltage and speed?
56. How do you scale down the input of an isolated converter?
57. Design a simple 4-bit DAC.
58. Explain the working principle of an ADC.
59. Explain the working principle of a buck converter.
60. Explain gain formulas.

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3. Power Electronics, Power Integrity & Power Delivery

1. Explain the different types of power supplies.
2. What are the pros and cons of different voltage regulators?
3. What is the difference between buck, boost, and buck-boost converters?
4. Explain the working algorithm of a buck-boost circuit.
5. How would you reduce 5 V to 3.3 V in a PCB circuit?
6. What is the response of an LDO under heavy load?
7. Describe LDO stability.
8. What is the difference between an LDO and a DC-DC converter?
9. Does an inductor source current or voltage in a buck converter?
10. What happens when an inductor in a DC-DC converter saturates?
11. Describe the components of a boost circuit you worked on.
12. What does each component in a boost circuit do?
13. How would you characterize the power efficiency of a power supply?
14. Explain power delivery voltage-current analysis.
15. What are the types of power supplies?

16. How would you measure PSRR?
17. Explain power scaling for IPs.
18. Explain power delivery fundamentals.
19. Explain voltage-current behavior in power circuits.
20. What hardware is used in data centres?
21. How would you design a battery-powered wearable system?
22. Explain system design for a battery-powered wearable.
23. How would you design a basic power circuit?
24. Explain power-related signal integrity concerns.
25. How would you place decoupling capacitors on power pins?

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4. Digital Design, VLSI, ASIC, FPGA & Verification

1. Convert a NAND gate into a NOT gate.
2. Draw a logic NAND gate.
3. What is the truth table of an XOR gate?
4. How would you implement an AND gate using a 2:1 multiplexer?
5. Realize factorial using logic gates.
6. Realize a power series using logic gates.
7. Explain flip-flops.
8. Explain the operation of a D flip-flop.
9. Derive the K-map for a JK flip-flop.
10. How would you implement a Q flip-flop using D flip-flops?
11. Explain setup time and hold time.
12. Explain setup and hold time violations.

13. Explain clock skew.
14. Explain timing on flip-flops.
15. Explain variation in semiconductor technology.
16. Explain asynchronous circuits.
17. Explain metastability.
18. How do you safely transfer a signal across different clock domains?
19. What are the risks if clock-domain crossing is not handled correctly?
20. Describe how a multi-bit synchronizer handles variable delay for each bit.
21. Describe how an asynchronous FIFO handles variable delay for each bit.
22. Explain CDC.
23. Explain RDC.
24. Explain FIFO depth.
25. Explain FIFO logic.
26. Explain FIFO buffers.
27. Explain verification methods for a parallel bus.
28. Explain SystemVerilog interfaces.
29. Explain blocking and non-blocking assignments.
30. Write Verilog code for an arbiter.
31. Explain finite state machines.
32. Build a state machine that outputs 1 when the serial input pattern is 1101.
33. Explain advanced FSM logic.
34. How would you improve Verilog code for power and area?
35. Explain physical design.
36. Explain hardware verification with computer architecture knowledge.

37. Explain full-chip verification.
38. Explain ASIC design and verification.
39. Explain digital electronics fundamentals.
40. Explain VLSI fundamentals.
41. Explain CMOS hardware questions.
42. Explain inverter layout.
43. Explain standard cells.
44. Given a standard-cell clock, generate a half-frequency clock.
45. Generate a one-third-frequency clock.
46. Generate a one-third-frequency clock with 50% duty cycle.
47. Design a frequency counter circuit.
48. Explain frequency dividers.
49. Explain cutoff frequency.
50. Explain time delay.
51. Explain semiconductor technology basics.
52. What is the architecture difference between a normal processor and a pipelined processor?
53. Explain pipelining.
54. What FPGA programming have you done?
55. Explain hardware description language experience.
56. Explain FPGA-related project work.
57. Explain Verilog language questions.
58. Explain design trade-offs in digital logic.
59. Explain cache-coherent design.
60. Explain FSM design trade-offs.

61. Explain digital design and signal integrity questions.
62. Explain silicon design concepts.
63. What is a TAP controller?
64. Explain JTAG/TAP-related hardware test concepts.
65. Explain DFT considerations when designing products.

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5. Computer Architecture, Memory & Processor Systems

1. Explain computer architecture concepts.
2. Explain memory and cache concepts.
3. What is a cache?
4. Explain cache replacement policies.
5. Which cache replacement policy is better? Show using memory accesses.
6. Explain cache address structures.
7. Explain cache hierarchy.
8. Explain virtual memory.
9. Explain virtual address concepts.
10. Explain register renaming.
11. Why do we need branch prediction?
12. Explain branch prediction.
13. Explain load/store bypassing and forwarding.
14. Explain memory coherence in multicore processors.
15. Explain the basic structure of a multicore processor.
16. Explain out-of-order superscalar processor structure.
17. Explain out-of-order execution trade-offs.

18. Explain CPU architecture.
19. Draw the internal architecture of an iPhone.
20. What is the difference between a microcontroller and a microprocessor?
21. What is a computer and a chip?
22. Explain computer architecture questions from your resume.
23. Explain hardware verification with computer architecture knowledge.
24. Explain system-level architecture.
25. Explain memory hardware fault detection.
26. How would you detect a hardware fault in memory using C?
27. Explain RAM testing.
28. Explain TAP controller knowledge.
29. Explain memory/cache-related digital electronics questions.
30. Explain processor pipeline architecture.

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6. Signal Integrity, PCB, Transmission Lines & Interfaces

1. What is the Nyquist criterion?
2. Why do we do impedance matching?
3. Explain transmission lines.
4. Explain signal integrity fundamentals.
5. Explain engineering fundamentals, test engineering, and signal integrity.
6. How would you change PCB tracks to maintain impedance if core spacing needed to double?
7. How much PCB experience do you have?
8. Explain PCB debugging.
9. Explain lab equipment, analog design, and op-amp-related PCB questions.

10. Explain basic PCB/component identification.
11. Identify electrical components, tools, and datasheets.
12. Name components on a motherboard.
13. Do you know how to read a schematic?
14. Explain the schematic provided during the interview.
15. Analyze a given schematic.
16. Analyze a given circuit.
17. What are the voltage waveforms in the given circuit?
18. Explain voltage regulation and inverter layout.
19. How would you place capacitors for power integrity?
20. How would you minimize CPU usage while collecting data from 1000 sensors?
21. Block-diagram a schematic to interface 1000 sensors and collect information periodically.
22. What are signal synchronization challenges across different clock domains?
23. Explain signal integrity issues in digital design.
24. Explain impedance matching in PCB routing.
25. Explain transmission-line behavior in hardware systems.
26. Explain USB, PCIe, SPI, I2C, and Ethernet fundamentals.
27. Explain how different communication protocols work.
28. What is UART?
29. What is I2C?
30. What is SPI?
31. Explain SPI, I2C, and Ethernet hardware debugging.
32. Explain basic telecommunication concepts.

33. What is the difference between CDMA and GSM?
34. What is MIMO, and what are its advantages?
35. What is the length of a GSM time slot?

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7. Sensors, Optics, Camera, RF & Wireless

1. What physiological factors can affect a PPG measurement?
2. What wavelength range of light can humans see?
3. Why is the sky blue?
4. What is UV space?
5. What is autofocus?
6. What are the main parts of a camera?
7. Explain camera measurement questions.
8. Explain hardware questions related to a camera.
9. Explain problem solving related to EE and cameras.
10. Explain optics alignment.
11. Explain physical optics.
12. Explain optical design.
13. Explain hardware control in LiDAR systems.
14. Explain time-of-flight measurement.
15. Explain image sensors.
16. Explain image-sensor characterization.
17. How would you remove crosstalk between channels in a CMOS image sensor?
18. Describe a sensor and the important parameters in its design.
19. What increases noise in a sensor?

20. How would you reduce noise in sensor readings?
21. Explain sensor experience in cars and aerospace applications.
22. Explain RF basics.
23. Explain noise figure.
24. Explain gain formulas.
25. How do you design an antenna for multiband operation?
26. What challenges come with multiband antenna design?
27. Why is there a null for a dipole antenna along its vertical axis?
28. Explain capacitive touch.
29. How else could you design a touch interface besides capacitive touch?
30. Explain capacitance sensing methods.
31. Explain magnetic hysteresis.
32. How do you define and measure magnetic hysteresis?
33. Explain audio fundamentals about amplifiers and measurements.
34. Explain robotics-related hardware experience.
35. What challenge have you faced in robotics, and how did you overcome it?

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8. Mechanical, Materials, Reliability & Physics

1. Explain the physics of vibrations.
2. Explain vibration controls.
3. Explain linear control, vibration, and sensing.
4. Explain beam theory.
5. Explain mechanical design concepts.
6. Explain mechanics of materials.

7. Draw a shear-force and bending-moment diagram.
8. What material is much stronger in compression than in tension?
9. What does strain mean?
10. Define strain.
11. What are the different components of the stress-strain curve?
12. Explain the behavior of hollow and non-hollow balls falling from a ramp.
13. Derive the differential equations of motion for a mechanical system.
14. Calculate the transfer function of a mechanical system.
15. When selecting material for a failed part, would you consider tensile strength or yield strength more?
16. Explain material-choice questions.
17. Explain advantages and disadvantages of tetrahedral and hexagonal FEA elements.
18. Explain shock testing.
19. Explain drop testing.
20. Give scenarios for shock and drop testing.
21. Explain reliability-related mechanical testing.
22. Explain product failure from a mechanical/material perspective.
23. How would you design a house?
24. Instead of hiring you as a hardware engineer, suppose you are asked to build a house. What questions would you ask before starting?
25. How would you plan a hang-gliding trip for colleagues to North Korea?

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9. Embedded Systems, Firmware, Programming & Scripting

1. Write code to convert a number to an Excel spreadsheet column header.

2. Do you have experience using a programming language not listed in the job requirement?
3. Write a C function to detect whether a number is odd.
4. Write a C function to count characters in a string.
5. Set up a simple Python class and work with its object.
6. Write a one-liner in Python to calculate the square of numbers in an array.
7. Write a swap function.
8. Write Fibonacci recursively and non-recursively.
9. Explain linked lists.
10. Code an in-place linked-list reverse.
11. Code an STL iterator.
12. Write a simple bash script.
13. How would you replace the word “Netflix” with “YouTube” in 1000 files using the command line?
14. Write C code to divide an 8-bit binary number by 4.
15. Find the equilibrium element in a string or list of numbers.
16. How would you randomly shuffle an array given a function that samples a random integer between 0 and n?
17. Explain space and time complexity.
18. Explain threading.
19. Explain heap and stack concepts.
20. Explain the difference between a semaphore and a mutex.
21. Explain programming-language questions from your resume.
22. What are the advantages and disadvantages of using Python for embedded systems development?
23. Explain embedded systems experience.

24. What are the different types of communication protocols?
25. How do the blocks in an embedded system interact with each other?
26. What are the challenges of implementing an embedded system?
27. How would you improve an embedded system?
28. Explain a simple coding problem from a hardware interview.
29. Write the code used in the Meltdown attack and explain how it works.
30. Explain C and Python coding/math project questions.
31. Explain simple C questions.
32. Explain queue design.
33. Explain the design of a queue.
34. Explain scripting experience.
35. Have you had experience with scripting?

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10. System Design, Product Design & Architecture Thinking

1. Describe system design for a battery-powered wearable.
2. Describe a system and explain how you would design it.
3. How would you start a project when little information is given?
4. How would you redesign part of your project given certain constraints?
5. How would you describe the engineering design process to a nontechnical person?
6. Explain your projects step by step.
7. Explain the purpose of each block in a block diagram.
8. How do blocks interact with each other in a system?
9. What are the challenges of implementing this system?
10. How would you improve this system?

11. Explain a system-design question at the end of the interview.
12. How would you design the original iPod scroll wheel?
13. Draw the internal architecture of an iPhone.
14. Design a house.
15. What questions would you ask before designing a house?
16. How would you design a touch interface besides capacitive touch?
17. How would you interface 1000 sensors and collect information periodically while minimizing CPU usage?
18. What test-design considerations would you use for a water bottle?
19. How would you make decisions during design?
20. What trade-offs would you consider in system design?
21. Explain product design and manufacturing operations questions.
22. Explain accessory design and coding questions.
23. Explain design questions involving hardware constraints.
24. How would you design a fixture for testing a subassembly?
25. How would you design a system using a 2-input/2-output block that outputs max and min values?
26. Using the max/min block, how would you build a 4-by-4 system that sorts or separates values from top to bottom?
27. How would you build a circuit/system based on limited requirements?
28. Come up with an interview question and answer it yourself.

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11. Math, Statistics, Logic & Brainteasers

1. What is standard deviation?
2. Explain basic linear algebra concepts.
3. How many times does the minute hand cross the hour hand in 24 hours?

4. You are in a fishing boat with a 1 kg rock. If you throw it to the bottom of the water, will the water level rise or fall?
5. Solve simple logic questions.
6. Solve brainteaser-style questions.
7. What colour do you get when you mix yellow and blue?
8. Why is the sky blue?
9. Explain bias and variance in machine learning.
10. Explain FIR and IIR filters.
11. Explain finite-element-analysis element choices.
12. Explain the mathematical transfer function of a mechanical system.
13. Calculate the resistance between two opposite corners of a resistor cube.
14. Find the equilibrium element in a string/list of numbers.
15. Explain basic models and calculations.
16. Answer number-theory-style questions.
17. AA > AE — what is the problem?
18. Explain how to divide numbers using logic systems.
19. Explain how to realize mathematical functions using logic gates.
20. Explain random shuffling using a random integer generator.

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12. Resume, Projects, Experience & Behavioral Questions

1. Tell me about yourself.
2. Tell me about your experience.
3. Describe your past experience.
4. Describe your last role.
5. Describe your current role.

6. Describe your role and what you did.
7. Walk through your resume.
8. Discuss the projects on your resume.
9. Explain your project briefly.
10. Explain your most recent project from your resume.
11. Describe your past projects in technical detail.
12. Explain your PhD project details.
13. Explain your master's thesis.
14. Tell me more about the internship you are doing.
15. What personal project have you worked on, and can you share it?
16. Talk about a project and answer follow-up questions.
17. Answer follow-up questions about your project presentation.
18. Explain questions based on your presentation.
19. Explain your research experience.
20. Introduce your research experience.
21. Give an example from your previous projects that demonstrates your professional expertise.
22. Describe previous work experience related to the role.
23. What relevant experience do you have for this position?
24. Walk through your experience against the job responsibilities.
25. Describe your QA or networking environment experience in 300 words or less.
26. Describe your experience with mechanical and software teams.
27. Talk about your robotics experience.
28. Describe your experience with sensors for cars and aerospace applications.

29. Have you worked with contract manufacturers?
30. Have you worked with CM/contract manufacturers?
31. How much PCB experience do you have?
32. What issue did you face in your role, and how did you resolve it with timelines?
33. Describe a difficult situation from your past experience and how you overcame it.
34. Tell me about a time you had a hard time at work and how you dealt with it.
35. Was there a time you did not get along with another team member? What happened, and how did you resolve it?
36. What is your biggest accomplishment?
37. What is your greatest weakness?
38. What best describes you as a person?
39. What are your best traits?
40. How would you improve overall team morale or motivation?
41. Why do you want this position?
42. Why are you interested in this particular position?
43. Why do you want this job?
44. Why do you want to work at Apple?
45. Why Apple?
46. What do you want to bring to Apple?
47. What do you know about the company?
48. Why do you want to work for Siemens eMobility?
49. In 200 words or less, explain why Vivint Smart Home interests you and how the internship aligns with your career plans.

50. Give me three reasons why I should hire you.
51. Why did you leave your previous company?
52. Do you need authorization to work in the US?
53. When do you graduate?
54. Are you willing to travel a lot for business trips?
55. Are you willing to relocate to a different country for this job?
56. If you could retake any class in college, what would it be?
57. Do you remember who has interviewed you so far? List their names.
58. Ask questions about the role, company culture, and mutual fit.
59. Answer STAR-method behavioral questions.
60. How do you start a project when little information is given? Answer using STAR logic.
61. How would you make decisions?
62. Explain family background and current living standard questions.
63. Describe your background and personal projects.
64. Explain technical challenges from your past.
65. Explain resume and background questions.
66. Explain questions related to previous experiences.
67. Explain role-fit questions based on the job description.

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13. Unclear, Vague, or Non-Disclosed Entries Kept for Completeness

These were present in the supplied content but did not contain enough technical detail to become a specific interview question.

1. "Engineering fundamentals, test engineering, SI."

2. “Hardware questions on CMOS.”
3. “Basic question about analog and digital circuit.”
4. “There were some systems and EE fundamental questions.”
5. “Basic circuit theory questions with different subjects from resume.”
6. “Pure technical and some technical challenges in past.”
7. “Resume.”
8. “Questions from your resume and background.”
9. “Questions about the project you did.”
10. “Questions were related to the job description and required detailed whiteboard explanation.”
11. “Physical design.”
12. “Ask for basic circuit design questions.”
13. “More technical questions.”
14. “Questions regarding the resume.”
15. “Topics put in my resume.”
16. “Nothing technical was asked.”
17. “Difficult.”
18. “Lab testing questions I never encountered before.”
19. “24-hour assignment for op-amps; questions confidential.”
20. “Not available due to confidentiality.”
21. “No particular question.”
22. “Standard interview questions.”
23. “Be well prepared for typical HR questions.”
24. “Basic models and calculation.”
25. “Technical questions from school texts.”

26. “Design questions on analog op-amp circuits.”
27. “Questions related to Verilog language.”
28. “Computer architecture, design verification, and system design.”
29. “Coding questions despite the role not requiring professional-level coding.”
30. “Technical test and hands-on test.”
31. “Written skills test.”
32. “General hardware-test questions.”
33. “Small/basic technical concepts.”
34. “Technical questions based on resume.”
35. “Behavioral questions, basic EE questions, and resume discussion.”

Clean category summary

Category	Main focus
Hardware Test & Debug	Test plans, lab setup, failure analysis, PCB/debug flow
Analog/Mixed-Signal	Op-amps, filters, ADC/DAC, transistors, circuit analysis
Power	LDO, DC-DC, buck/boost, PSRR, power efficiency
Digital/VLSI	FSM, CDC, FIFO, Verilog, DFT, timing, CMOS
Computer Architecture	Cache, virtual memory, pipelining, OoO, branch prediction
PCB/SI/Interfaces	Transmission lines, impedance, UART/I2C/SPI, schematics
Sensors/RF/Optics	PPG, image sensors, antennas, LiDAR, RF, MIMO
Mechanical/Materials	Vibrations, stress-strain, beams, shock/drop, FEA
Coding/Scripting	C, Python, bash, data structures,

Category	Main focus
	complexity
System Design	Wearables, sensor systems, product design, architecture
Behavioral/Resume	Experience, projects, role fit, STAR questions
Vague/Unclear	Topic-only or non-disclosed entries preserved separately

Is this conversation helpful so far?